# HospitAI

## From raw electronic health records to readable clinical insight

A hospital analytics platform that groups patients by clinical profile, mines the care pathways inside each group, and explains both in plain language. Built end to end on the synthetic **Synthea** dataset, with no real patient data.


## Todor Ioan-Iulian

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### ANALYTICAL PIPELINE

1. **Synthea EHR** — 22k synthetic patients, 10+ tables
2. **Feature building** — 21 features / patient, JSONB
3. **K-Prototypes** — mixed numeric + categorical clustering
4. **PrefixSpan** — sequence mining, per cluster
5. **LLM insight** — cluster summaries + text-to-SQL

**Stack** Next.js · FastAPI · PostgreSQL
**ML** kmodes · prefixspan · LangGraph
**Roles** patient · doctor · manager

## Dataset & Features

Real EHRs are GDPR-locked, so **Synthea** generated a synthetic cohort with deep longitudinal histories across 10+ relational tables (encounters, conditions, medications, procedures, immunizations, allergies, observations, care plans, providers, organizations).

| 22,284 | 10+ | 21 |
|---|---|---|
| patients | tables | features |

- An ETL job loads the CSVs into **PostgreSQL**; a feature-builder aggregates every table into **one row per patient**.
- **11 numeric** features: age, BMI, systolic BP, active conditions, medications, procedures, ER visits, QALY, DALY.
- **10 categorical** features: gender, BMI type, condition burden, diabetes / hypertension / heart-disease flags, dominant care plan.
- Cached as a **JSONB** column so clustering and mining never recompute it.

Synthea · PostgreSQL · FastAPI ETL

## K-Prototypes Clustering

**K-Prototypes** (Huang, 1998) extends K-Means to **mixed data**: squared-Euclidean distance for numerics plus matching dissimilarity for categoricals, combined in one distance.

- An IQR rule keeps **21,995** patients for model fitting; the **289** high-acuity outliers are retained as a separate scope.
- Benchmarked **63 feature versions** × a sweep over scaler, γ, seed, threshold and **k ∈ {4..7}**, scored on 6 cluster-quality indices.
- The **Gower silhouette** is weighted highest, the only index that honours the mixed metric; FAMD projections act as tie-breakers.
- Final pick **v55 (k=4)**: Gower **0.350**, with all four clusters above the 0.20 cohesion heuristic.
- Top drivers: active conditions ($\eta^2$=0.74), age (0.66); condition burden (V=0.80), diabetes (0.69).

mixed data · Gower silhouette · FAMD 2-D

## Sequence Mining

Each patient's history becomes **one chronological sequence** of events typed by prefix (**C:**condition, **M:**medication, **P:**procedure, **CP:**care plan). **PrefixSpan** (Pei et al., 2001) mines frequent ordered sub-sequences by **prefix-projection**, growing a pattern one event at a time and recursing only on the suffixes that follow, so it never enumerates candidates. It runs **once per cluster**, so each phenotype reveals its own trajectory.

- Sequences are cleaned first: noise blacklisted, refills collapsed, a **730-day window** and length cap applied.
- **Closed** patterns are ranked by **geometric-mean lift**, which normalises by each item's baseline frequency so rare, meaningful chains beat high-support trivia.
- A **type-diversity bonus** favours condition → medication → procedure chains; **Jaccard 0.80** dedup and dental caps remove the rest, then survivors are attributed back to their patients.

```
# a mined trajectory is a typed, ordered chain
C:Prediabetes → M:Metformin → P:HbA1c measurement
→ P:Diabetic retinal exam
```

## AI Interpretation

Large language models turn the platform's statistical output into language that non-technical staff can read and act on, bridging numbers to decisions.

- **Cluster summaries:** each cluster's statistics become a **title**, a narrative description, a **risk level** and a list of recommended interventions.
- **Text-to-SQL agent** on **LangGraph**: a question such as *"patients with >3 emergencies this year"* becomes SQL, is executed, and is answered in natural language.
- The generated SQL stays **visible** as a transparency safeguard, so staff can verify exactly what ran against the data.
- Outputs are strictly **descriptive**, never medical advice or a substitute for a clinician.
- Three roles (**patient**, **doctor**, **manager**) each see a tailored slice of the same data.

LangGraph · Gemini / GPT · text-to-SQL

## Results

**k=4** is the smallest partition where all four clusters clear the 0.20 cohesion bar, giving a clean morbidity gradient from a healthy young majority to a small, high-burden outlier pool. The deciding metric was the **Gower silhouette** (the only index honouring the mixed numeric/categorical distance), picked among several indices of inter-cluster separation and intra-cluster cohesion.

**c0 · young / low-burden** — **silhouette 0.423**
Acute bronchitis → Respiratory therapy → Acetaminophen **432 pts**

**c1 · reproductive women** — **silhouette 0.199**
Normal pregnancy → Pregnancy test → Fetal-viability ultrasound **432 pts**

**c2 · hypertensive adults** — **silhouette 0.320**
Furosemide IV → Oxygen by mask → Prone positioning (heart failure) **164 pts**

**c3 · pediatric / acute** — **silhouette 0.396**
Otitis media → Amoxicillin → Ibuprofen **320 pts**

**−1 · high-acuity outliers** — **289 patients**
Chemoradiation → Cisplatin → Paclitaxel → Hospice care **132 pts**

## Key Findings & Impact

1. **The cohorts are real, not artefacts.** Mixed clustering splits the hospital into four recognisable phenotypes along a morbidity gradient, each cohesive enough to clear the silhouette bar.
2. **Each cohort tells its own story.** Mining per cluster turns every phenotype into a signature trajectory (a child's ear infection, an antenatal work-up, a heart-failure rescue) that a single global run would average away.
3. **Ranking by lift makes them readable.** Geometric-mean lift rewards rare, meaningful chains and suppresses Synthea's 40%-support dental noise, so what surfaces is clinical, not bureaucratic.
4. **The rarest patients matter most.** Mining the 289 outliers as their own scope exposes the high-cost oncology and renal chains that fall below any global support floor.
5. **And it reaches people.** Clinicians see where a patient fits and the pathway their cohort follows; managers query the hospital in plain language, served fast from offline batch jobs persisted to PostgreSQL.